

Nonparametric Regression

Regression models are used to study how a dependent or response variable depends on an independent variable or variables. The regression function $f(x)$ is defined as the mean for a response variable Y when the independent variable equals x . In model form, with n observations, we may write

$$Y_i = f(x_i) + \epsilon_i, \quad i = 1, \dots, n,$$

with $E\epsilon_i = 0$.

Classically, the regression function f is assumed to lie in a class of functions specified by a finite number of parameters. For instance, in quadratic regression f is a quadratic function, $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$, specified by three parameters, β_0 , β_1 , and β_2 . This approach feels natural with a small- or moderate-size data set, as the data in this case may not be rich enough to support fitting a more complicated model. But with more data a researcher will often want to consider more involved models, since in most applications there is little reason to believe the regression function lies exactly in some narrow parametric family. Of course one could add complexity by increasing the number of parameters, fitting perhaps a cubic or quartic function, say, instead of a quadratic. But this approach may have limitations, and recently there has been considerable interest in replacing parametric assumptions about f with more qualitative assumptions about the smoothness of f .

In this chapter we explore two approaches. We begin with kernel methods, based on Clark (1977), which exploit the assumed smoothness of f in a fairly direct fashion. With this approach the regression function is estimated as a weighted average of the responses with similar values for the independent variable. The other approach, splines, is derived by viewing the regression function f as an unknown parameter taking values in an infinite-dimensional vector space. This approach is developed in Section 18.3, following a section extending results about finite-dimensional vector spaces to Hilbert spaces. The chapter closes with a section showing how similar ideas can be used for density estimation.

18.1 Kernel Methods

Consider a regression model in which

$$Y_i = f(x_i) + \epsilon_i, \quad i = 1, \dots, n,$$

where $\epsilon_1, \dots, \epsilon_n$ are mean zero, uncorrelated random variables with a common variance σ^2 . The independent variables $x_1, \dots, x_n$ are viewed as (observed or known) constants, and the response variables $Y_1, \dots, Y_n$ are observed and random. The errors $\epsilon_1, \dots, \epsilon_n$ are not observed, and σ^2 is an unobserved parameter. The regression function f is unknown and is not assumed to lie in any parametric class. But we do assume it is twice continuously differentiable. Finally, for convenience, assume $x_1 < \dots < x_n$.

One conceivable estimator for f might be the function $\hat{h}$ obtained from the data by linear interpolation. This function is Y_i when $x = x_i$ and is linear between adjacent values for x , so

$$\hat{h}(x) = Y_i \frac{x - x_{i+1}}{x_i - x_{i+1}} + Y_{i+1} \frac{x - x_i}{x_{i+1} - x_i}, \quad x \in [x_i, x_{i+1}]. \quad (18.1)$$

If the errors are very small then $\hat{h}$ may lie close to f , but when the errors are appreciable $\hat{h}$ will jump up and down too much to be a sensible estimator. This can be seen in the plot to the left in Figure 18.1, with the true regression function f shown as a dashed line and $\hat{h}$ given as a solid line.

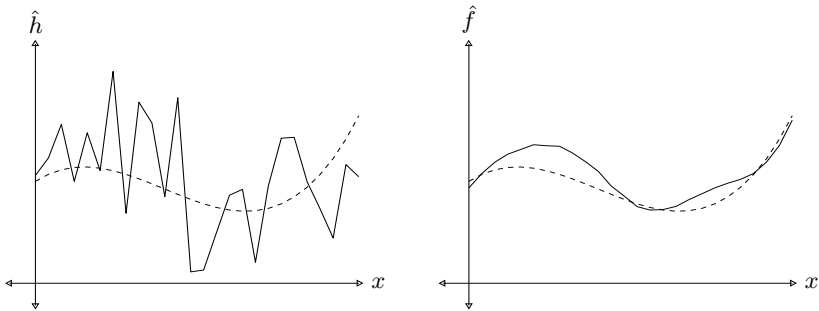


Fig. 18.1. Kernel smoothing: left: $\hat{h}$; right: $\hat{f}$.

One way to make a function smoother is through convolution. Doing this with $\hat{h}$ leads to an estimator

$$\hat{f}(x) = \frac{1}{b} \int \hat{h}(t) W\left(\frac{x-t}{b}\right) dt, \quad (18.2)$$

where W is a probability density and b , called the *bandwidth*, controls the amount of smoothing. The plot to the right in Figure 18.1 shows¹ $\hat{f}$, and it is

¹ The graphs are based on simulated data with $f(x) = 16x^3 - 19.2x^2 + 4.68x + 2.3$, $\epsilon_i \sim N(0, 1)$, $W(x) = (1 - |x|)^+$, $b = 0.24$, and $x_{i+1} - x_i = 0.04$. To avoid problems with edge effects, $\hat{f}$ is based on observations (not graphed) with $x \notin [0, 1]$.

indeed smoother than $\hat{h}$. Viewing this integral in (18.2) as an expectation,

$$\hat{f}(x) = E_Z \hat{h}(x - bZ),$$

where Z is an absolutely continuous variable with density W . From this we can see that $\hat{f}(x)$ is a weighted average of $\hat{h}(y)$ over values y of order b from x . With increasing b there is more averaging and, as intuition suggests, this will improve the variance of $\hat{f}(x)$. But if f curves near x , averaging will induce bias in the estimator $\hat{f}$ that grows as b increases. We explore this below when the independent variables are equally spaced, trying to choose the bandwidth to balance these concerns.

The estimators $\hat{h}$ and $\hat{f}$ are both linear functions of Y . Using (18.1),

$$\hat{h}(x) = \sum_{i=1}^n u_i(x) Y_i,$$

where

$$u_i(x) = \begin{cases} \frac{x - x_{i-1}}{x_i - x_{i-1}}, & x \in [x_{i-1}, x_i]; \\ \frac{x - x_{i+1}}{x_i - x_{i+1}}, & x \in [x_i, x_{i+1}]; \\ 0, & \text{otherwise.} \end{cases}$$

Using this in (18.2),

$$\hat{f}(x) = \frac{1}{b} \int \sum_{i=1}^n u_i(t) Y_i W\left(\frac{x-t}{b}\right) dt = \sum_{i=1}^n v_i(x) Y_i, \quad (18.3)$$

where

$$v_i(x) = \frac{1}{b} \int u_i(t) W\left(\frac{x-t}{b}\right) dt. \quad (18.4)$$

With this linear structure, moments of $\hat{f}(x)$ should be easy to compute.

If we let $h(x) = E\hat{h}(x)$, then by (18.1),

$$h(x) = f(x_i) \frac{x - x_{i+1}}{x_i - x_{i+1}} + f(x_{i+1}) \frac{x - x_i}{x_{i+1} - x_i}, \quad x \in [x_i, x_{i+1}],$$

so h is the linear interpolant of f . The difference between h and f can be estimated by Taylor approximation. If $x \in [x_i, x_{i+1}]$, then

$$f(x) = f(x_i) + (x - x_i)f'(x_i) + \int_{x_i}^x (x - u)f''(u) du.$$

In particular, if $x = x_{i+1}$, we have

$$f(x_{i+1}) = f(x_i) + (x_{i+1} - x_i)f'(x_i) + \int_{x_i}^{x_{i+1}} (x_{i+1} - u)f''(u) du,$$

and we can use this to eliminate $f'(x_i)$ from the first equation in favor of $f(x_{i+1})$. The algebra is messy but straightforward and gives

$$\begin{aligned} f(x) &= f(x_i) + (x - x_i) \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} \\ &\quad - \int_{x_i}^x (u - x_i) \left(1 - \frac{x - x_i}{x_{i+1} - x_i} \right) f''(u) du \\ &\quad - \int_x^{x_{i+1}} (x - x_i) \frac{x_{i+1} - u}{x_{i+1} - x_i} f''(u) du \\ &= h(x) - \frac{1}{2}(x - x_i)(x_{i+1} - x) \int_{x_i}^{x_{i+1}} p_{i,x}(u) f''(u) du, \end{aligned}$$

where $p_{i,x}$ is a probability density concentrated on (x_i, x_{i+1}) . If f'' is continuous, then by the (first) intermediate value theorem for integrals,

$$f(x) = h(x) - \frac{1}{2}(x - x_i)(x_{i+1} - x) f''(u_{i,x}),$$

where $u_{i,x} \in (x_i, x_{i+1})$. In particular, since $(x - x_i)(x_{i+1} - x) \leq \frac{1}{4}(x_{i+1} - x_i)^2$, if $M_2 = \sup |f''(x)|$, then

$$|f(x) - h(x)| \leq \frac{M_2}{8}(x_{i+1} - x_i)^2. \quad (18.5)$$

For clarity, let us now assume that the x_i are equally spaced between 0 and 1, $x_i = i/n$. Assume also that W is continuous, symmetric, and has support $[-1, 1]$. Two popular choices for the kernel are $W_2(x) = (1 - |x|)^+$ and $W_4(x) = \frac{3}{4}(1 - x^2)^+$. Then

$$\begin{aligned} Ef(x) &= E \frac{1}{b} \int \hat{h}(t) W\left(\frac{x-t}{b}\right) dt \\ &= \frac{1}{b} \int E \hat{h}(t) W\left(\frac{x-t}{b}\right) dt \\ &= \frac{1}{b} \int f(t) W\left(\frac{x-t}{b}\right) dt + \frac{1}{b} \int (h(t) - f(t)) W\left(\frac{x-t}{b}\right) dt. \end{aligned}$$

By (18.5), the difference between f and h is at most $M_2/(8n^2)$, and since W is a probability density integrating to one, this also bounds the magnitude of the final term.² If Z has density W , then the other term is

$$EZf(x - bZ) = EZ[f(x) - bZf'(x) + \frac{1}{2}b^2Z^2f''(x - bZ^*)],$$

where Z^* is an intermediate value in $(-1, 1)$. Because W is symmetric, $EZ = 0$, and by dominated convergence

² We neglect “edge effects” here, assuming $x_i > b$ and $x_{i+1} < 1 - b$.

$$\text{Bias}(\hat{f}(x)) = E\hat{f}(x) - f(x) = \frac{1}{2}b^2 f''(x)EZ^2 + o(b^2) + O(1/n^2),$$

as $b \rightarrow 0$. With the regularity imposed, the asymptotics here also hold uniformly in x provided we stay away from the endpoints 0 and 1.

From the representation (18.3),

$$\text{Var}(\hat{f}(x)) = \sigma^2 \sum_{i=1}^n v_i^2(x).$$

Note that in this sum, the number of nonzero terms is of order nb because $v_i(x)$ is zero unless $|x - x_i| \leq b + 1/n$. To approximate the terms in this sum, note that nu_i is a probability measure concentrated on (x_{i-1}, x_{i+1}) , and so, by (18.4),

$$v_i(x) = \frac{1}{nb} W\left(\frac{x - t_i^*(x)}{b}\right),$$

with $t_i^*(x)$ some value in (x_{i-1}, x_{i+1}) . If $M_1 = \sup |W'(t)|$, then

$$\left| v_i(x) - \frac{1}{nb} W\left(\frac{x - x_i}{b}\right) \right| \leq \frac{M_1}{n^2 b^2}.$$

Since the points x_i/b are uniformly spaced and separated by an amount $1/(nb)$, in a limit in which $b \rightarrow 0$ but $nb \rightarrow \infty$, then

$$\frac{1}{nb} \sum_{i=1}^n W^2\left(\frac{x - x_i}{b}\right)$$

is a Riemann approximation for $\int W^2(x-t) dt$ converging to $\int W^2(t) dt$. Thus

$$\begin{aligned} \text{Var}(\hat{f}(x)) &= \frac{\sigma^2}{nb} \frac{1}{nb} \sum_{i=1}^n \left[W^2\left(\frac{x - x_i}{b}\right) + O\left(\frac{1}{nb}\right) \right] \\ &= \frac{\sigma^2}{nb} \left[\int W^2(t) dt + o(1) \right] + O\left(\frac{1}{n^2 b^2}\right) \\ &\sim \frac{\sigma^2}{nb} \int W^2(t) dt. \end{aligned}$$

Combining our approximations for the bias and variance of $\hat{f}(x)$, we can approximate the mean square error of $\hat{f}(x)$ as

$$\begin{aligned} \text{MSE}(x) &= E[\hat{f}(x) - f(x)]^2 = \text{Var}(\hat{f}(x)) + \text{Bias}^2(\hat{f}(x)) \\ &\approx \frac{\sigma^2}{nb} \int W^2(t) dt + b^4 \frac{1}{4} (f''(x))^2 (EZ^2)^2. \end{aligned}$$

The mean square error measures the performance of $\hat{f}$ at individual values for the independent variable x . For a more global assessment, the integrated mean square error may be a natural measure:

$$\begin{aligned}\text{IMSE} &= \int_0^1 \text{MSE}(x) dx \\ &= \int_0^1 E[\hat{f}(x) - f(x)]^2 dx = E \int_0^1 [\hat{f}(x) - f(x)]^2 dx.\end{aligned}$$

Using the approximation for the mean square error,

$$\text{IMSE} \approx \frac{c_1}{nb} + b^4 c_2,$$

where

$$c_1 = \sigma^2 \int W^2(t) dt \quad \text{and} \quad c_2 = \frac{1}{4}(EZ^2)^2 \int_0^1 (f''(x))^2 dx.$$

The approximation for the integrated mean square error, viewed as a function of b , is minimized at a value where the derivative is zero; that is, at a value solving

$$-\frac{c_1}{nb^2} + 4c_2 b^3 = 0.$$

This gives

$$b_{opt} = \left(\frac{c_1}{4nc_2} \right)^{1/5}$$

as an optimal choice (approximately), and with this choice

$$\text{IMSE} \approx 5c_2^{1/5} \left(\frac{c_1}{4n} \right)^{4/5} = K_f \left(\frac{\sigma^2}{n} \right)^{4/5} \left(\left[\int W^2(t) dt \right]^4 [EZ^2]^2 \right)^{1/5},$$

where

$$K_f = \frac{5}{4} \left(\int_0^1 (f''(x))^2 dx \right)^{1/5},$$

which depends on f but is independent of the kernel W . Using W_2 , $\text{IMSE} \approx 0.353K_f(\sigma^2/n)^{4/5}$, and using W_4 , $\text{IMSE} \approx 0.349K_f(\sigma^2/n)^{4/5}$. Thus W_4 has a slight theoretical advantage.

In practice, the choice of the kernel is less important than the choice of the bandwidth. The formula for b_{opt} cannot be used directly, since the constants c_1 and c_2 depend on σ^2 and $\int_0^1 (f''(x))^2 dx$. One natural idea is to estimate these quantities somehow and choose the bandwidth by plugging the estimates into the formula for b_{opt} . This is feasible, but a bit tricky since derivatives of f are often harder to estimate than f itself. Another idea would be to use a *cross-validation* approach based on prediction error. Suppose we wanted to predict the outcome at a new location x . The expected squared prediction error would be

$$E[Y - \hat{f}(x)]^2 = E[f(x) + \epsilon - \hat{f}(x)]^2 = \sigma^2 + \text{MSE}(x).$$

If x were chosen at random from a uniform distribution on $(0, 1)$, then the expected squared prediction error would be $\sigma^2 + \text{IMSE}$. Here is a resampling method to estimate this error:

1. Partition the data at random into two samples, an estimation sample with n_1 observations and a validation sample with n_2 observations. Let $\hat{f}_b^*$ denote the kernel estimate for f based on the estimation sample with bandwidth b .
2. Define

$$C(b) = \frac{1}{n_2} \sum_{i=1}^{n_2} (y_{2,i} - \hat{f}_b^*(x_{2,i}))^2,$$

where $(x_{2,i}, y_{2,i})$, $i = 1, \dots, n_2$, are the data in the validation sample.

3. Repeat steps 1 and 2 m times and define

$$\overline{C}(b) = \frac{1}{m} \sum_{i=1}^m C_i(b),$$

where $C_i(b)$ is the function $C(b)$ for the i th partition.

The value $\hat{b}$ minimizing $\overline{C}$ would be the cross-validation choice for the bandwidth.

18.2 Hilbert Spaces

If V is a vector space in $\mathbb{R}^n$ with an orthonormal basis $e_1, \dots, e_p$, then any $x \in V$ can be written as

$$x = \sum_{i=1}^p c_i e_i,$$

with the constants c_i in the expansion given by

$$c_i = e_i' x, \quad i = 1, \dots, p.$$

Classes of functions may also form vector spaces over $\mathbb{R}$, but these spaces are rarely spanned by a finite set of functions. However, expansions like those above may be possible with an infinite collection of basis vectors. To deal with infinite sums we need a notion of convergence, and this is based here on the norm or length of a vector. And, for the geometric structure of interest, we also need inner products. Here are formal definitions of norms and inner products.

Definition 18.1. *Let V be a vector space over $\mathbb{R}$. A norm on V is a real-valued function on V , $\|\cdot\| : V \rightarrow \mathbb{R}$, satisfying the following conditions:*

1. *For all x in V , $\|x\| \geq 0$, and $\|x\| = 0$ only if $x = \mathbb{O}$ (the zero vector in V).*
2. *For all x and y in V , $\|x + y\| \leq \|x\| + \|y\|$.*
3. *For all $x \in V$ and $c \in \mathbb{R}$, $\|cx\| = |c| \times \|x\|$.*

Using a norm we can define convergence $x_n \rightarrow x$ to mean that $\|x_n - x\| \rightarrow 0$. Then a function f from one normed space to another is *continuous* if $f(x_n) \rightarrow f(x)$ whenever $x_n \rightarrow x$. For instance the function $f(x) = \|x\|$ is continuous. The second property of norms implies

$$\|x_n\| \leq \|x_n - x\| + \|x\| \quad \text{and} \quad \|x\| \leq \|x - x_n\| + \|x_n\|.$$

Together these imply

$$|f(x_n) - f(x)| = |\|x_n\| - \|x\|| \leq \|x_n - x\|,$$

which tends to zero (by definition) whenever $x_n \rightarrow x$.

Definition 18.2. Let V be a vector space over $\mathbb{R}$. An inner product is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ that is symmetric:

$$\langle x, y \rangle = \langle y, x \rangle, \quad \forall y, x \in V;$$

bilinear:

$$\langle x, ay + bz \rangle = a\langle x, y \rangle + b\langle x, z \rangle$$

and

$$\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle,$$

for all x, y, z in V and all a, b in $\mathbb{R}$; and positive definite:

$$\langle x, x \rangle \geq 0,$$

with equality only if $x = \mathbb{O}$. The pair $(V, \langle \cdot, \cdot \rangle)$ is called an inner product space.

Proposition 18.3. In an inner product space, $\|x\| = \sqrt{\langle x, x \rangle}$ defines a norm satisfying the Cauchy-Schwarz inequality,

$$|\langle x, y \rangle| \leq \|x\| \times \|y\|. \quad (18.6)$$

Proof. The first property of a norm follows because the inner product is positive definite, the third property follows from the bilinearity which gives $\langle cx, cx \rangle = c^2 \langle x, x \rangle$, and, anticipating (18.6), the second property of a norm follows because

$$\begin{aligned} |\langle x + y, x + y \rangle| &= |\langle x, x \rangle + 2\langle x, y \rangle + \langle y, y \rangle| \\ &\leq \|x\|^2 + 2\|x\| \times \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2. \end{aligned}$$

To finish we must verify the Cauchy-Schwarz inequality. It is not hard to show that $\langle x, \mathbb{O} \rangle = 0$, and so the inequality is immediate unless x and y are both nonzero. In this case,

$$\langle x - cy, x - cy \rangle = \|x\|^2 - 2c\langle x, y \rangle + c^2\|y\|^2,$$

viewed as a function of c , is minimized when $c = \langle x, y \rangle / \|y\|^2$. But the function is nonnegative for all c , and so plugging in the minimizing value we have

$$\|x\|^2 - 2\frac{\langle x, y \rangle^2}{\|y\|^2} + \frac{\langle x, y \rangle^2}{\|y\|^2} \geq 0.$$

After a bit of rearrangement this gives (18.6). $\square$

One consequence of the Cauchy–Schwarz inequality is that the inner product $\langle \cdot, \cdot \rangle$ is continuous, because

$$\begin{aligned} |\langle \tilde{x}, \tilde{y} \rangle - \langle x, y \rangle| &= |\langle \tilde{x}, \tilde{y} - y \rangle + \langle \tilde{x} - x, y \rangle| \\ &\leq \|\tilde{x}\| \times \|\tilde{y} - y\| + \|\tilde{x} - x\| \times \|y\|, \end{aligned}$$

which tends to zero as $\tilde{x} \rightarrow x$ and $\tilde{y} \rightarrow y$.

If a norm $\|\cdot\|$ comes from an inner product, then

$$\|x \pm y\|^2 = \|x\|^2 \pm 2\langle x, y \rangle + \|y\|^2.$$

Adding these two relations we have the *parallelogram law*, stating that

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2. \quad (18.7)$$

Elements x and y in an inner product space V are called *orthogonal*, written $x \perp y$, if $\langle x, y \rangle = 0$. Since

$$\langle x + y, x + y \rangle = \langle x, x \rangle + 2\langle x, y \rangle + \langle y, y \rangle,$$

we then have the Pythagorean relation,

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

If W is a subspace of V , then the orthogonal complement of W is

$$W^\perp = \{x \in V : \langle x, y \rangle = 0, \forall y \in W\}.$$

If $x_n \rightarrow x$, then $\|x_n - x\| \rightarrow 0$, which implies that

$$\sup_{m:m \geq n} \|x_m - x\| \rightarrow 0,$$

as $n \rightarrow \infty$. Because

$$\|x_n - x_m\| \leq \|x_n - x\| + \|x - x_m\|,$$

convergence implies

$$\lim_{n \rightarrow \infty} \sup_{m \geq n} \|x_n - x_m\| = 0. \quad (18.8)$$

Sequences satisfying this equation are called *Cauchy*, but if the space is not rich enough some Cauchy sequences may not converge. For instance, 3, 3.1, 3.14, ... is a Cauchy sequence in $\mathbb{Q}$ without a limit in $\mathbb{Q}$, because π is irrational.

Definition 18.4. A normed vector space V is complete if every Cauchy sequence in V has a limit in V . A complete inner product space is called a Hilbert space.

The next result extends our notion of projections in Euclidean spaces to Hilbert spaces.

Theorem 18.5. Let V be a closed subspace of a Hilbert space $\mathcal{H}$. For any $x \in \mathcal{H}$ there is a unique $y \in V$, called the projection of x onto V , minimizing $\|x - z\|$ over $z \in V$. Then $x - y \in V^\perp$, and this characterizes y : if $\tilde{y} \in V$ and $x - \tilde{y} \in V^\perp$, then $\tilde{y} = y$.

Proof. Let $d = \inf_{z \in V} \|x - z\|$ (the distance from x to V), and choose y_n in V so that $\|x - y_n\| \rightarrow d$. By the parallelogram law,

$$\|x - y_m + x - y_n\|^2 + \|y_n - y_m\|^2 = 2\|x - y_n\|^2 + 2\|x - y_m\|^2.$$

But

$$\|2x - y_m - y_n\|^2 = 4\left\|x - \frac{1}{2}(y_n + y_m)\right\|^2 \geq 4d^2,$$

and so

$$\|y_n - y_m\|^2 \leq 2\|x - y_n\|^2 + 2\|x - y_m\|^2 - 4d^2 \rightarrow 0,$$

as $m, n \rightarrow \infty$. So y_n , $n \geq 1$, is a Cauchy sequence converging to some element $y \in \mathcal{H}$. Since V is closed, $y \in V$, and by continuity, $\|x - y\| = d$. Next, suppose $\tilde{y} \in V$ and $\|x - \tilde{y}\| = d$. If $z \in V$ then $\tilde{y} + cz \in V$ for all $c \in \mathbb{R}$ and

$$0 \leq \|x - \tilde{y} - cz\|^2 - \|x - \tilde{y}\|^2 = -2c\langle x - \tilde{y}, z \rangle + c^2\|z\|^2.$$

This can only hold for all $c \in \mathbb{R}$ if $\langle x - \tilde{y}, z \rangle = 0$, and thus $x - \tilde{y} \in V^\perp$. Finally, since y and $\tilde{y}$ both lie in V and $x - y$ and $x - \tilde{y}$ both lie in $V^\perp$,

$$\|y - \tilde{y}\|^2 = \langle x - \tilde{y}, y - \tilde{y} \rangle - \langle x - y, y - \tilde{y} \rangle = 0,$$

showing that y is unique. □

If V is a closed subspace of a Hilbert space $\mathcal{H}$, let $P_V x$ denote the projection of x onto V . The following result shows that P_V is a linear operator with operator norm one (see Problem 18.4).

Proposition 18.6. If V is a closed subspace of a Hilbert space $\mathcal{H}$, $x \in \mathcal{H}$, $y \in \mathcal{H}$, and $c \in \mathbb{R}$, then

$$P_V(cx + y) = cP_V x + P_V y$$

and

$$\|P_V x\| \leq \|x\|.$$

Proof. Since V is a subspace and P_Vx and P_Vy lie in V , $cP_Vx + P_Vy \in V$; and if $z \in V$,

$$\langle cx + y - cP_Vx - P_Vy, z \rangle = c\langle x - P_Vx, z \rangle + \langle y - P_Vy, z \rangle = 0,$$

because $x - P_Vx \perp z$ and $y - P_Vy \perp z$. Using Theorem 18.5, $cP_Vx + P_Vy$ must be the projection of $cx + y$ onto V . For the second assertion, since $P_Vx \in V$ and $x - P_Vx \in V^\perp$ are orthogonal, by the Pythagorean relation

$$\|P_Vx\|^2 + \|x - P_Vx\|^2 = \|x\|^2. \quad \square$$

Definition 18.7. A collection e_t , $t \in T$, is said to be orthonormal if $e_s \perp e_t$ for all $s \neq t$ and $\|e_t\| = 1$, for all t .

As in the finite-dimensional case, we would like to represent elements in our Hilbert space as linear combinations of elements in an orthonormal collection, but extra care is necessary because some infinite linear combinations may not make sense.

Definition 18.8. The linear span of $S \subset \mathcal{H}$, denoted $\text{span}(S)$, is the collection of all finite linear combinations $c_1x_1 + \cdots + c_nx_n$ with $c_1, \dots, c_n$ in $\mathbb{R}$ and $x_1, \dots, x_n$ in S . The closure of this set is denoted $\overline{\text{span}}(S)$.

Definition 18.9. An orthonormal collection e_t , $t \in T$, is called an orthonormal basis for a Hilbert space $\mathcal{H}$ if $\langle e_t, x \rangle \neq 0$ for some $t \in T$, for every nonzero $x \in \mathcal{H}$.

Theorem 18.10. Every Hilbert space has an orthonormal basis.

The proof in general relies on the axiom of choice. (The collection of all orthonormal families is inductively ordered, so a maximal element exists by Zorn's lemma, and any maximal element is an orthonormal basis.) When $\mathcal{H}$ is separable, a basis can be found by applying the Gram-Schmidt algorithm to a countable dense set, and in this case the basis will be countable.

Theorem 18.11. If e_n , $n \geq 1$, is an orthonormal basis, then each $x \in \mathcal{H}$ may be written as

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k.$$

Proof. Let

$$x_n = \sum_{k=1}^n \langle x, e_k \rangle e_k.$$

The infinite sum in the theorem is the limit of these partial sums, so we begin by showing that these partial sums form a Cauchy sequence. If $j \leq n$,

$$\langle x - x_n, e_j \rangle = \langle x, e_j \rangle - \sum_{k=1}^n \langle x, e_k \rangle \langle e_k, e_j \rangle = 0,$$

since $\langle e_k, e_j \rangle = 0$, unless $k = j$, and in that case it is 1. From this, $x - x_n \in \text{span}\{e_1, \dots, e_n\}^\perp$, and by Theorem 18.5, x_n is the projection of x onto $\text{span}\{e_1, \dots, e_n\}$. By the Pythagorean relation,

$$\|x_n\|^2 = \sum_{k=1}^n \langle x, e_k \rangle^2,$$

and by Proposition 18.6, $\|x_n\| \leq \|x\|$. From this we have Bessel's inequality,

$$\sum_{k=1}^n \langle x, e_k \rangle^2 \leq \|x\|^2,$$

and since n here is arbitrary, the coefficients $\langle x, e_k \rangle$, $k \geq 1$, are square summable. By the Pythagorean relation, if $n < m$,

$$\|x_m - x_n\|^2 = \left\| \sum_{k=n+1}^m \langle x, e_k \rangle e_k \right\|^2 = \sum_{k=n+1}^m \langle x, e_k \rangle^2,$$

which tends to zero as m and n tend to infinity. So x_n , $n \geq 1$, is a Cauchy sequence, and since $\mathcal{H}$ is complete the sequence must have a limit x_∞ . Because the inner product $\langle \cdot, \cdot \rangle$ is a continuous function of its arguments, for any $j \geq 1$,

$$\langle x - x_\infty, e_j \rangle = \lim_{n \rightarrow \infty} \langle x - x_n, e_j \rangle.$$

But if $n \geq j$, $\langle x - x_n, e_j \rangle = 0$ because x_n is the projection of x onto $\text{span}\{e_1, \dots, e_n\}$, and so the limit in this expression must be zero. Therefore

$$\langle x - x_\infty, e_j \rangle = 0, \quad j \geq 1.$$

Finally, since e_k , $k \geq 1$, form an orthonormal basis, $x - x_\infty$ must be zero, proving the theorem. $\square$

18.3 Splines

Let us consider again our nonparametric regression model

$$Y_i = f(x_i) + \epsilon_i, \quad i = 1, \dots, n,$$

where $\epsilon_1, \dots, \epsilon_n$ are mean zero, uncorrelated random variables with a common variance σ^2 . As with the kernel approach, there is a presumption that f is smooth. The smoothing spline approach tries to take direct advantage of this

smoothness by augmenting the usual least squares criteria with a penalty for roughness. For instance, if the x_i lie in $[0, 1]$, the estimator $\hat{f}$ might be chosen to minimize

$$J(f) = \sum_{i=1}^n (Y_i - f(x_i))^2 + \lambda \|f^{(m)}\|_2^2, \quad (18.9)$$

where $\|\cdot\|_2$ is the $\mathcal{L}_2$ -norm of functions on $[0, 1]$ under Lebesgue measure,

$$\|g\|_2^2 = \int_0^1 g^2(x) dx.$$

The constant λ is called a *smoothing parameter*. Larger values for λ will lead to a smoother $\hat{f}$, smaller values will lead to an estimate $\hat{f}$ that follows the observed data more closely, that is, with $\hat{f}(x_i)$ closer to Y_i .

For the roughness penalty in our criteria to make sense, $f^{(m-1)}$ will need to be absolutely continuous according to the following definition.

Definition 18.12. A real-valued function g defined on an interval of $\mathbb{R}$ is absolutely continuous if there exists a function g' such that

$$g(b) - g(a) = \int_a^b g'(x) dx, \quad \forall a < b.$$

If g is differentiable, then g' must be the derivative a.e., so use of a common notation should not cause any confusion. Also, if f has $m-1$ continuous derivatives and $g = f^{(m-1)}$ is absolutely continuous, then we denote g' as $f^{(m)}$.

Definition 18.13. The Sobolev space $W_m[0, 1]$ is the collection of all functions $f : [0, 1] \rightarrow \mathbb{R}$ with $m-1$ continuous derivatives, $f^{(m-1)}$ absolutely continuous, and $\|f^{(m)}\|_2 < \infty$. With an inner product $\langle \cdot, \cdot \rangle$ defined by

$$\langle f, g \rangle = \sum_{k=0}^{m-1} f^{(k)}(0)g^{(k)}(0) + \int_0^1 f^{(m)}(x)g^{(m)}(x) dx, \quad f, g \in W_m[0, 1],$$

$W_m[0, 1]$ is a Hilbert space.

These Hilbert spaces have an interesting structure. Suppose we define

$$K(x, y) = \sum_{k=0}^{m-1} \frac{1}{k!^2} x^k y^k + \int_0^{x \wedge y} \frac{(x-u)^{m-1}(y-u)^{m-1}}{(m-1)!^2} du.$$

Then

$$\frac{\partial^k}{\partial x^k} K(x, y) \Big|_{x=0} = \frac{y^k}{k!}, \quad k = 0, \dots, m-1,$$

and

$$\frac{\partial^m}{\partial x^m} K(x, y) = \frac{(y-x)^{m-1}}{(m-1)!} I\{x \leq y\}. \quad (18.10)$$

Comparing this with the Taylor expansion

$$f(y) = \sum_{k=0}^{m-1} \frac{1}{k!} f^{(k)}(0) y^k + \int_0^y \frac{(y-x)^{m-1} f^{(m)}(x)}{(m-1)!} dx,$$

we see that

$$f(y) = \langle f, K(\cdot, y) \rangle.$$

This formula shows that the evaluation functional, $f \rightsquigarrow f(y)$, is a bounded linear operator. Hilbert spaces in which this happens are called *reproducing kernel Hilbert spaces*. The function K here is called the reproducing kernel, reproducing because

$$K(x, y) = \langle K(\cdot, x), K(\cdot, y) \rangle.$$

The kernel K is a positive definite function. To see this, first note that

$$\sum_{i,j} c_i c_j K(x_i, x_j) = \sum_{i,j} c_i c_j \langle K(\cdot, x_i), K(\cdot, x_j) \rangle = \left\| \sum_i c_i K(\cdot, x_i) \right\|^2,$$

which is nonnegative. If this expression is zero, then $h = \sum_i c_i K(\cdot, x_i)$ is zero. But then $\langle h, f \rangle = \sum_i c_i f(x_i)$ will be zero for all f , which can only happen if $c_i = 0$ for all i .

To minimize $J(f)$ in (18.9) over $f \in W_m[0, 1]$, let Π_m denote the vector space of all polynomials of degree at most $m-1$, let $\eta_i = K(\cdot, x_i)$, $i = 1, \dots, n$, and define

$$V = \Pi_m \oplus \text{span}\{\eta_1, \dots, \eta_n\}.$$

An arbitrary function f in $W_m[0, 1]$ can be written as $g + h$ with $g \in V$ and $h \in V^\perp$. Because h is orthogonal to η_i , $h(x_i) = \langle h, \eta_i \rangle = 0$. Also, if $k \leq m-1$, then the inner product of h with the monomial x^k is $k!h^{(k)}(0)$, and because h is orthogonal to these monomials, $h^{(k)}(0) = 0$, $k = 0, \dots, m-1$. It follows that $\|h\| = \|h^{(m)}\|_2$, and

$$\langle g, h \rangle = \int_0^1 g^{(m)}(x) h^{(m)}(x) dx = 0.$$

But then

$$\|g^{(m)} + h^{(m)}\|_2^2 = \int_0^1 (g^{(m)}(x) + h^{(m)}(x))^2 dx = \|g^{(m)}\|_2^2 + \|h^{(m)}\|_2^2,$$

and so

$$\begin{aligned}
 J(f) &= J(g+h) = \sum_{i=1}^n (Y_i - g(x_i) - h(x_i))^2 + \lambda \|g^{(m)} + h^{(m)}\|_2^2 \\
 &= \sum_{i=1}^n (Y_i - g(x_i))^2 + \lambda \|g^{(m)}\|_2^2 + \lambda \|h^{(m)}\|_2^2 = J(g) + \lambda \|h\|^2.
 \end{aligned}$$

From this it is evident that a function minimizing J must lie in V .

Using (18.10),

$$\eta_i^{(m)}(x) = \frac{(x_i - x)^{m-1}}{(m-1)!} I\{x \leq x_i\}.$$

From this, on $[0, x_i]$, η_i must be a polynomial of degree $2m-1$, and on $[x_i, 1]$, η_i is a polynomial of degree at most $m-1$. Taking more derivatives,

$$\eta_i^{(m+j)}(x) = \frac{(-1)^j (x_i - x)^{m-1-j}}{(m-1-j)!} I\{x \leq x_i\}, \quad j = 1, \dots, m-2,$$

and so the derivatives of η_i of order $2m-2$ or less are continuous. Linear combinations of the η_i are piecewise polynomials. Functions like these are called *splines*.

Definition 18.14. A function $f : [0, 1] \rightarrow \mathbb{R}$ is called a spline of order q with (simple) knots $0 < x_1 < \dots < x_n < 1$ if, for any $i = 0, \dots, n$, the restriction of f to $[x_i, x_{i+1}]$ (with the convention $x_0 = 0$ and $x_{n+1} = 1$) is a polynomial of degree $q-1$ or less, and if the first $q-2$ derivatives of f are continuous on the whole domain $[0, 1]$. The collection of all splines of order q is denoted $\mathcal{S}_q = \mathcal{S}_q(x_0, \dots, x_{n+1})$. The space $\mathcal{S}_q$ is a vector space.

From the discussion above, any function $f \in V$ must be a spline of order $2m$. In addition, all functions $f \in V$ are polynomials of degree $m-1$ on the last interval $[x_n, 1]$. So if $\hat{f}$ minimizes $J(f)$, it will be a polynomial of degree at most $m-1$ on $[x_n, 1]$, and by time reversal³ it will also be a polynomial of degree $m-1$ or less on the first interval $[0, x_1]$. It will then be a *natural spline* according to the following definition.

Definition 18.15. A function $f \in \mathcal{S}_{2q}$ is called a natural spline of order $2q$ if its restrictions to the first and last intervals, $[0, x_1]$ and $[x_n, 1]$, are polynomials of degree $q-1$ or less. Let $\hat{\mathcal{S}}_{2q}$ denote the set of all natural splines.

These spline spaces (with fixed knots) are finite-dimensional vector spaces, and once we know that the function $\hat{f}$ minimizing J lies in a finite-dimensional vector space, $\hat{f}$ can be identified using ordinary linear algebra. To see how, let e_j , $j = 1, \dots, k$, be linearly independent functions with

³ Formally, the argument just given shows that the function $\tilde{f}(t) \stackrel{\text{def}}{=} \hat{f}(1-t)$, that minimizes $\sum (Y_i - f(1-x_i))^2 + \lambda \|f^{(m)}\|_2^2$, must be a polynomial of degree at most $m-1$ on $[1-x_1, 1]$.

$$\tilde{\mathcal{S}}_{2m} \subset \text{span}\{e_1, \dots, e_k\}.$$

If

$$f = c_1 e_1 + \dots + c_k e_k,$$

then

$$f(x_i) = \langle f, \eta_i \rangle = \sum_j c_j \langle e_j, \eta_i \rangle = [Ac]_i,$$

where A is a matrix with entries

$$A_{ij} = \langle e_j, \eta_i \rangle, \quad i = 1, \dots, n, \quad j = 1, \dots, k.$$

Then

$$\sum_{i=1}^n (Y_i - f(x_i))^2 = \|Y - Ac\|^2 = Y'Y - 2Y'Ac + c'A'Ac.$$

For the other term in J ,

$$\|f^{(m)}\|_2^2 = \left\| \sum_{j=1}^k c_j e_j^{(m)} \right\|_2^2 = \sum_{i,j} c_i c_j \int_0^1 e_i^{(m)}(x) e_j^{(m)}(x) dx = c' B c,$$

where B is a $k \times k$ matrix with

$$B_{ij} = \int_0^1 e_i^{(m)}(x) e_j^{(m)}(x) dx, \quad i = 1, \dots, k, \quad j = 1, \dots, k.$$

Using these formulas,

$$J(f) = Y'Y - 2Y'Ac + c'A'Ac + \lambda c'Bc.$$

This is a quadratic function of c with gradient

$$2(A'A + \lambda B)c - 2A'Y.$$

Setting the gradient to zero, if

$$\hat{c} = (A'A + \lambda B)^{-1} A'Y$$

and

$$\hat{f} = \sum_{j=1}^k \hat{c}_j e_j,$$

then $\hat{f}$ minimizes $J(f)$ over $W_m[0, 1]$.

One collection of linearly independent functions with span containing⁴ $\tilde{\mathcal{S}}_{2q}$ is given by

⁴ For some linear combinations of these functions, restriction to the final interval $[x_n, 1]$ will give a polynomial of degree greater than $q - 1$. So the linear span of these functions is in fact strictly larger than $\tilde{\mathcal{S}}_{2q}$.

$$e_i(x) = (x - x_i)_+^{2q-1}, \quad i = 1, \dots, n,$$

along with the monomials of degree $q - 1$ or less,

$$e_{n+j}(x) = x^{j-1}, \quad j = 1, \dots, q.$$

This can be seen recursively. If $f \in \tilde{S}_{2q}$, let

$$p = \sum_{j=1}^q c_j e_{n+j}$$

be a polynomial of degree $q - 1$ equal to f on $[0, x_1]$. Then $f - p \in \tilde{S}_{2q}$ is zero on $[0, x_1]$. By the enforced smoothness for derivatives at the knots, on $[x_1, x_2]$ $f - p$ will be a polynomial of degree $2q - 1$ with $2q - 2$ derivatives equal to zero at x_1 . Accordingly, on this interval $f - p = c_1(x - x_1)^{2q-1}$, and it follows that $f - p - c_1 e_1$ is zero on $[0, x_2]$. Next, with a proper choice of c_2 , $f - p - c_1 e_1 - c_2 e_2$ will be zero on $[0, x_3]$. Further iteration eventually gives $f - p - \sum_{j=1}^n c_j e_j = 0$ on $[0, 1]$.

The choice of the spanning functions $e_1, \dots, e_k$ may not seem important from a mathematical perspective. But a careful choice can lead to more efficient numerical algorithms. For instance, $\tilde{S}_2$ contains the functions

$$e_1(x) = \begin{cases} 1, & x \in [0, x_1]; \\ \frac{x_2 - x}{x_2 - x_1}, & x \in [x_1, x_2]; \\ 0, & \text{otherwise,} \end{cases}$$

$$e_n(x) = \begin{cases} \frac{x - x_{n-1}}{x_n - x_{n-1}}, & x \in [x_{n-1}, x_n]; \\ 1, & x \in [x_n, 1]; \\ 0, & \text{otherwise,} \end{cases}$$

and

$$e_i(x) = \begin{cases} \frac{x - x_{i-1}}{x_i - x_{i-1}}, & x \in [x_{i-1}, x_i]; \\ \frac{x_{i+1} - x}{x_{i+1} - x_i}, & x \in [x_i, x_{i+1}]; \\ 0, & \text{otherwise,} \end{cases} \quad i = 2, \dots, n-1.$$

These functions are called *B-splines*. The first two, e_1 and e_2 , are plotted in [Figure 18.2](#). The *B-splines* are linearly independent and form what is called a *local basis* for $\tilde{S}_2$, because each basis function vanishes except on at most two adjacent intervals of the partition induced by $x_1, \dots, x_n$. Functions expressed in this basis can be computed quickly at a point x since all but at most two terms in $\sum c_j e_j(x)$ will be zero. In addition, with this basis, the matrix A is the identity, and B will be “tridiagonal,” with $B_{ij} = 0$ if $|i - j| \geq 2$. Matrices

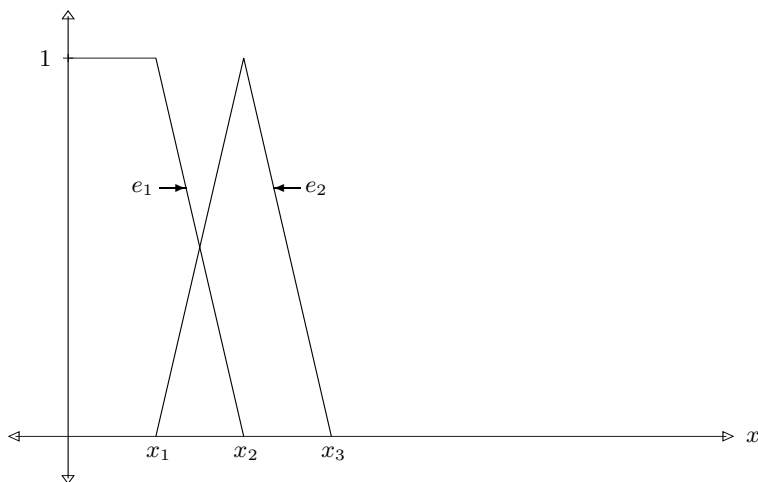


Fig. 18.2. B -splines e_1 and e_2 .

with a banded structure can be inverted and multiplied much more rapidly than matrices with arbitrary entries.

B -spline bases are also available for other spline spaces. The notation needed to define them carefully is a bit involved, but it is not too hard to understand why they should exist. In $\mathcal{S}_q$, the $q + 1$ functions

$$(x - x_i)_+^{q-1}, \dots, (x - x_{i+q})_+^{q-1}$$

restricted to $[x_{i+q}, 1]$ are all polynomials of degree $q - 1$. Because polynomials of degree $q - 1$ form a vector space of dimension q , the restrictions cannot be linearly dependent, and some nontrivial linear combination of these functions must be zero on $[x_{i+q}, 1]$. This linear combination gives a function in $\mathcal{S}_q$ that is zero unless its argument lies in (x_i, x_{i+q}) . With a suitable normalization, functions such as this form a B -spline basis for $\mathcal{S}_q$. For further information on splines see Wahba (1990) or De Boor (2001)

18.4 Density Estimation

The methods just developed for nonparametric regression can also be applied to nonparametric density estimation. Let $X_1, \dots, X_n$ be i.i.d. from some distribution Q . One natural estimator for Q would be the *empirical distribution* $\hat{Q}$ defined by

$$\hat{Q}(A) = \frac{1}{n} \# \{i \leq n : X_i \in A\}.$$

This estimator $\hat{Q}(\cdot)$ is a discrete distribution placing atoms with mass $1/n$ at each observation X_i . So integrals against $\hat{Q}_n$ are just averages,

$$\int g d\hat{Q}_n = \frac{1}{n} \sum_{i=1}^n g(X_i).$$

If we believe Q is absolutely continuous with a smooth density f , $\hat{Q}$ is not a very sensible estimator; it is too rough, in much the same way the linear interpolant $\hat{h}$ of the data was too rough for estimating a smooth regression function. A kernel approach to estimating f uses convolution, as in Problem 18.10, to smooth $\hat{Q}$. Intuition suggests this may give a reasonable estimate for f if the convolving distribution is concentrated near zero. To accomplish this we incorporate a bandwidth b , tending to zero as $n \rightarrow \infty$, and consider estimators of the form

$$\hat{f}(x) = \frac{1}{b} \int W\left(\frac{x-t}{b}\right) d\hat{Q}(t) = \frac{1}{nb} \sum_{i=1}^n W\left(\frac{x-X_i}{b}\right),$$

with W a fixed symmetric probability density.

With the linear structure, formulas for the mean and variance for $\hat{f}(x)$ are easy to derive and study. If f'' is continuous and bounded, then

$$E\hat{f}(x) = f(x) + \frac{1}{2}b^2 f''(x) \int t^2 W(t) dt + o(b^2) \quad (18.11)$$

and

$$\text{Var}(\hat{f}(x)) = \frac{1}{nb} f(x) \|W\|_2^2 + o(1/(nb)) \quad (18.12)$$

uniformly in n as $b \downarrow 0$. Combining these, the mean square error for $\hat{f}(x)$ is

$$\begin{aligned} \text{MSE}(x) &= E(\hat{f}(x) - f(x))^2 \\ &= \frac{f(x)}{nb} \|W\|_2^2 + \frac{1}{4}b^4 (f''(x))^2 \left(\int t^2 W(t) dt \right)^2 + o(b^4 + 1/(nb)) \end{aligned}$$

as $b \downarrow 0$. If $b = b_n$ varies with n so that $b_n \rightarrow 0$ and $nb_n \rightarrow \infty$, then this mean square error will tend to zero.

With suitable regularity, this approximation can be integrated, giving

$$\begin{aligned} \text{IMSE} &= E \int (\hat{f}(x) - f(x))^2 dx = \int \text{MSE}(x) dx \\ &= \frac{1}{nb} \|W\|_2^2 + \frac{1}{4}b^4 \|f''\|_2^2 \left(\int t^2 W(t) dt \right)^2 + o(b^4 + 1/(nb)). \end{aligned}$$

Minimizing this approximation,

$$b \sim \frac{\|W\|_2^{2/5}}{n^{1/5} \|f''\|_2^{2/5} \left(\int t^2 W(t) dt \right)^{2/5}}$$

will be asymptotically optimal, and with this choice

$$\text{IMSE} \sim \frac{5}{4} \|W\|_2^{8/5} \|f''\|_2^{2/5} \left(\int t^2 W(t) dt \right)^{2/5} n^{-4/5}.$$

For further discussion, see Chapter 2 of Wand and Jones (1995)

A spline approach to density estimation is more challenging. If we assume $f > 0$ and take $\theta = \log f$, then an estimator $\hat{f} = e^{\hat{\theta}}$ for f will automatically be positive. For regularity, let us assume $\theta \in W_m[0, 1]$. In contrast to nonparametric regression, since f must integrate to one, θ cannot vary freely over $W_m[0, 1]$, but must satisfy the constraint

$$\int_0^1 e^{\theta(x)} dx = 1.$$

Let Ω denote the class of all functions in $W_m[0, 1]$ satisfying this constraint. The log-likelihood function is given by

$$l(\theta) = \sum_{i=1}^n \theta(X_i).$$

A direct maximum likelihood approach to estimating θ or f fails because

$$\sup_{\theta \in \Omega} l(\theta) = \infty,$$

with arbitrarily high values for the likelihood achieved by densities with very large spikes at the data values. To mitigate this problem, we incorporate a penalty for smoothness and choose $\hat{\theta}$ to maximize

$$J_0(\theta) = \frac{1}{n} \sum_{i=1}^n \theta(X_i) - \lambda \|\theta^{(m)}\|_2^2.$$

To ameliorate troubles with the constraint, Silverman (1982) introduces another functional,

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n \theta(X_i) - \int_0^1 e^{\theta(x)} dx - \lambda \|\theta^{(m)}\|_2^2.$$

Theorem 18.16. *The function $\hat{\theta} \in \Omega$ maximizes J_0 over Ω if and only if $\hat{\theta}$ maximizes J over $W_m[0, 1]$.*

Proof. If $\theta \in W_m[0, 1]$ and $c = \int_0^1 e^{\theta(x)} dx$, then $\tilde{\theta} = \theta - \log c \in \Omega$ and $\|\theta^{(m)}\|_2 = \|\tilde{\theta}^{(m)}\|_2$. Thus

$$J(\tilde{\theta}) = J(\theta) - \log c - 1 + c.$$

But $c - \log c \geq 1$ for all $c > 0$, with equality only if $c = 1$, and thus $J(\tilde{\theta}) \geq J(\theta)$, with equality only if $c = 1$, that is, only if $\theta \in \Omega$. So any θ maximizing J over $W_m[0, 1]$ must lie in Ω . But on Ω , $J = J_0 - 1$, and the theorem follows. $\square$

The *null family* associated with the smoothness penalty here is defined as

$$\Omega_0 = \{\theta \in \Omega : \|\theta^{(m)}\|_2 = 0\}.$$

Functions in Ω_0 must be polynomials and can be parameterized as

$$\theta_\eta(x) = \eta_1 x + \cdots + \eta_{m-1} x^{m-1} - A(\eta),$$

where

$$A(\eta) = \log \int_0^1 \exp \left[\sum_{i=1}^{m-1} \eta_i x^i \right] dx.$$

Then $\Omega_0 = \{\theta_\eta : \eta \in \mathbb{R}^{m-1}\}$ and the corresponding densities, $f_\eta = e^{\theta_\eta}$, $\eta \in \mathbb{R}^{m-1}$, form an exponential family.

If λ is large, then functions $\theta \in \Omega$ that are not close to Ω_0 will incur a substantial smoothness penalty. For this reason, if $\hat{\theta}_\lambda$ is the estimator maximizing J , then $\hat{\theta}_\lambda$ should converge as $\lambda \rightarrow \infty$ to the maximum likelihood estimator for the null family Ω_0 . Let us call this estimator $\hat{\theta}_\infty$. In applications, this observation might be used in a reverse fashion to choose a smoothness penalty. If there is reason to believe that the data come from some particular exponential family, a researcher may want to choose a penalty with these target distributions as its null family. The next result shows that existence is also tied to estimation for the null family.

Theorem 18.17. *With a given data set $X_1, \dots, X_n$, J will have a maximizer in Ω if $\hat{\theta}_\infty$, the maximum likelihood estimator for the null family, exists. This will hold with probability one if $n \geq m$.*

Proposition 18.18. *The functional J is strictly concave.*

Proof. Given θ_1, θ_2 in $W_m[0, 1]$ and $\alpha \in (0, 1)$, since the exponential function is convex,

$$\begin{aligned} \int_0^1 e^{\alpha\theta_1(x) + (1-\alpha)\theta_2(x)} dx &\leq \int_0^1 \left[\alpha e^{\theta_1(x)} + (1-\alpha) e^{\theta_2(x)} \right] dx \\ &= \alpha \int_0^1 e^{\theta_1(x)} dx + (1-\alpha) \int_0^1 e^{\theta_2(x)} dx, \end{aligned}$$

with equality only if $\theta_1 = \theta_2$. Also,

$$\begin{aligned} \|\alpha\theta_1^{(m)} + (1-\alpha)\theta_2^{(m)}\|_2^2 &\leq [\alpha\|\theta_1^{(m)}\|_2 + (1-\alpha)\|\theta_2^{(m)}\|_2]^2 \\ &\leq \alpha\|\theta_1^{(m)}\|_2^2 + (1-\alpha)\|\theta_2^{(m)}\|_2^2. \end{aligned}$$

So

$$J(\alpha\theta_1 + (1-\alpha)\theta_2) \geq \alpha J(\theta_1) + (1-\alpha)J(\theta_2),$$

with equality if and only if $\theta_1 = \theta_2$. □

As a consequence of this result, if an estimator $\hat{\theta}$ maximizing J exists, it must be unique, for if $\hat{\theta}_1$ and $\hat{\theta}_2$ both maximize J , then

$$J(\hat{\theta}_1) = \frac{1}{2}J(\hat{\theta}_1) + \frac{1}{2}J(\hat{\theta}_2) \leq J\left(\frac{\hat{\theta}_1 + \hat{\theta}_2}{2}\right) \leq J(\hat{\theta}_1).$$

The inequalities here must be equalities, and strict concavity then implies $\hat{\theta}_1 = \hat{\theta}_2$.

The following result, from Silverman (1982), shows that with a suitable choice for the penalty scale λ , the estimator $\hat{\theta}$ is consistent.

Theorem 18.19. *Suppose $\theta \in W_{2m}[0, 1]$ and*

$$\theta^{(2m-1)}(0) = \theta^{(2m-1)}(1).$$

If $\lambda \rightarrow 0$ and $n^{m-\delta}\lambda \rightarrow \infty$ for some $\delta > 0$, then for every $\epsilon > 0$,

$$\|\hat{\theta} - \theta\|_{\infty}^2 = O_p\{\lambda^{-\epsilon}(n^{-1}\lambda^{-1/m} + \lambda^{(4m-1)/(2m)})\}.$$

In particular, if $\lambda = n^{-(2m)/(4m+1)}$,

$$\|\hat{\theta} - \theta\|_{\infty}^2 = O_p\left(n^{-(4m-1)/(4m+1)+\epsilon}\right),$$

for every $\epsilon > 0$.

18.5 Problems

1. *Estimating σ^2 .* Consider nonparametric regression with the assumptions in Section 18.1 and i.i.d. errors ϵ_i . A natural estimator for σ^2 might be

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{f}(x_i))^2,$$

with $\hat{f}$ the kernel estimator for f . Suppose $b = cn^{-1/5}$. Is this estimator necessarily consistent? Prove or explain why not. In your argument you can ignore edge effects.

2. *Locally weighted regression.* Like kernel smoothing, locally weighted regression is a linear approach to nonparametric regression. Let W be a continuous, nonnegative, symmetric ($W(x) = W(-x)$) function, decreasing on $(0, \infty)$ with support $[-1, 1]$. The estimate for f at a point x is based on weighted least squares, fitting a polynomial to the data with weights emphasizing data with x_i near x . This problem considers quadratic models in which $\hat{\beta}$ is chosen to minimize

$$\sum_{i=1}^n W\left(\frac{x_i - x}{b}\right) (y_i - \beta_1 - \beta_2(x_i - x) - \beta_3(x_i - x)^2)^2.$$

The estimate for $f(x)$ is then $\hat{f}(x) = \hat{\beta}_1$. Here the bandwidth b is taken to be a small constant (decreasing as n increases), although in practice b is often chosen using the x_i so that the estimate for $f(x)$ is based on a fixed number of data points. For simplicity, you may assume below that $x_i = i/n$.

- a) Derive an explicit formula for $\hat{f}(x_j)$ when $x_j \in (b, 1 - b)$.
 - b) Derive approximations for the bias of $\hat{f}(x_j)$ as $n \rightarrow \infty$ and $b \downarrow 0$ with $nb \rightarrow \infty$ and $x_j = \lfloor nx \rfloor / n$, $x \in (0, 1)$.
 - c) Derive an approximation for the variance of $\hat{f}(x_j)$ in the same limit.
 - d) What choice for bandwidth b minimizes the mean square error of $\hat{f}(x_j)$ (approximately).
3. Show that the stated inner product $\langle \cdot, \cdot \rangle$ in Definition 18.13 for $W_m[0, 1]$ satisfies the conditions in Definition 18.2
 4. Let $\mathcal{X}$ and $\mathcal{Y}$ be normed vector spaces over $\mathbb{R}$. A function $T : \mathcal{X} \rightarrow \mathcal{Y}$ is called a *linear operator* if

$$T(cx_1 + x_2) = cT(x_1) + T(x_2), \quad \forall x_1, x_2 \in \mathcal{X}, c \in \mathbb{R}.$$

The *operator norm* (or spectral norm) of T is defined as

$$\|T\| = \sup\{\|T(x)\| : \|x\| \leq 1\},$$

and T is called *bounded* if $\|T\| < \infty$.

- a) Show that a bounded operator T is continuous: If $\|x_n - x\| \rightarrow 0$, then $\|T(x_n) - T(x)\| \rightarrow 0$.
 - b) Show that a continuous linear operator T is bounded.
 - c) Let $\mathcal{X} = \mathbb{R}^m$ and $\mathcal{Y} = \mathbb{R}^n$, with the usual Euclidean norms. Let A be an $n \times m$ matrix, and define a linear operator T by $T(x) = Ax$. Relate the operator norm $\|T\|$ to the eigenvalues of $A'A$.
5. Consider the set $\mathcal{C}[0, 1]$ of continuous real functions on $[0, 1]$ with the $\mathcal{L}_2$ inner product

$$\langle x, y \rangle = \int_0^1 x(t)y(t) dt$$

and associated norm

$$\|x\|_2 = \sqrt{\int_0^1 x^2(t) dt}.$$

- a) Find a Cauchy sequence x_n (for this norm) that does not converge, showing that $\mathcal{C}[0, 1]$ is not complete (with this norm). (The usual norm for $\mathcal{C}[0, 1]$ is $\|x\| = \sup_{t \in (0, 1)} |x(t)|$, and with this norm $\mathcal{C}[0, 1]$ is complete.)

- b) Let T_t be the evaluation operator, $T_t(x) = x(t)$. Show that T_t is an unbounded linear operator.
6. Show that if $f_n \rightarrow f$ in $W_2[0, 1]$, then $f'_n(x) \rightarrow f'(x)$.
7. Find a nontrivial function $f \in \mathcal{S}_4(0, 0.1, 0.2, \dots, 1)$ which is zero unless its argument lies between 0.2 and 0.6.
8. Suppose $m = 1$, $x = (0.2, 0.4, 0.6, 0.8)$, and we use the B -spline basis described for $\tilde{\mathcal{S}}_2$. Calculate the matrices A and B that arise in the formula to compute $\hat{f} = \sum \hat{c}_j e_j$.
9. *Semiparametric models.* Consider a regression model with two explanatory variables x and w in which

$$Y_i = f(x_i) + \beta w_i + \epsilon_i, \quad i = 1, \dots, n,$$

with $0 < x_1 < \dots < x_n < 1$, $f \in W_m[0, 1]$, $\beta \in \mathbb{R}$, and the ϵ_i i.i.d. from $N(0, \sigma^2)$. This might be called a semiparametric model because the dependence on w is modeled parametrically, but the dependence on x is nonparametric. Following a penalized least squares approach, consider choosing $\hat{f}$ and $\hat{\beta}$ to minimize

$$J(f, \beta) = \sum_{i=1}^n (Y_i - f(x_i) - \beta w_i)^2 + \lambda \|f^{(m)}\|_2^2.$$

- a) Show that the estimator $\hat{f}$ will still be a natural spline of order $2m$.
- b) Derive explicit formulas based on linear algebra to compute $\hat{\beta}$ and $\hat{f}$.
10. *Convolutions.* Suppose $X \sim Q$ and $Y \sim W$ are independent, and that W is absolutely continuous with Lebesgue density w . Show that $T = X + Y$ is absolutely continuous with density h given by

$$h(t) = \int w(t - x) dQ(x).$$

The distribution of T is called the convolution of Q with W , and this shows that if either Q or W is absolutely continuous, their convolution is absolutely continuous.

11. Use dominated convergence to prove (18.11).
12. Prove (18.12).